CC LAB 8

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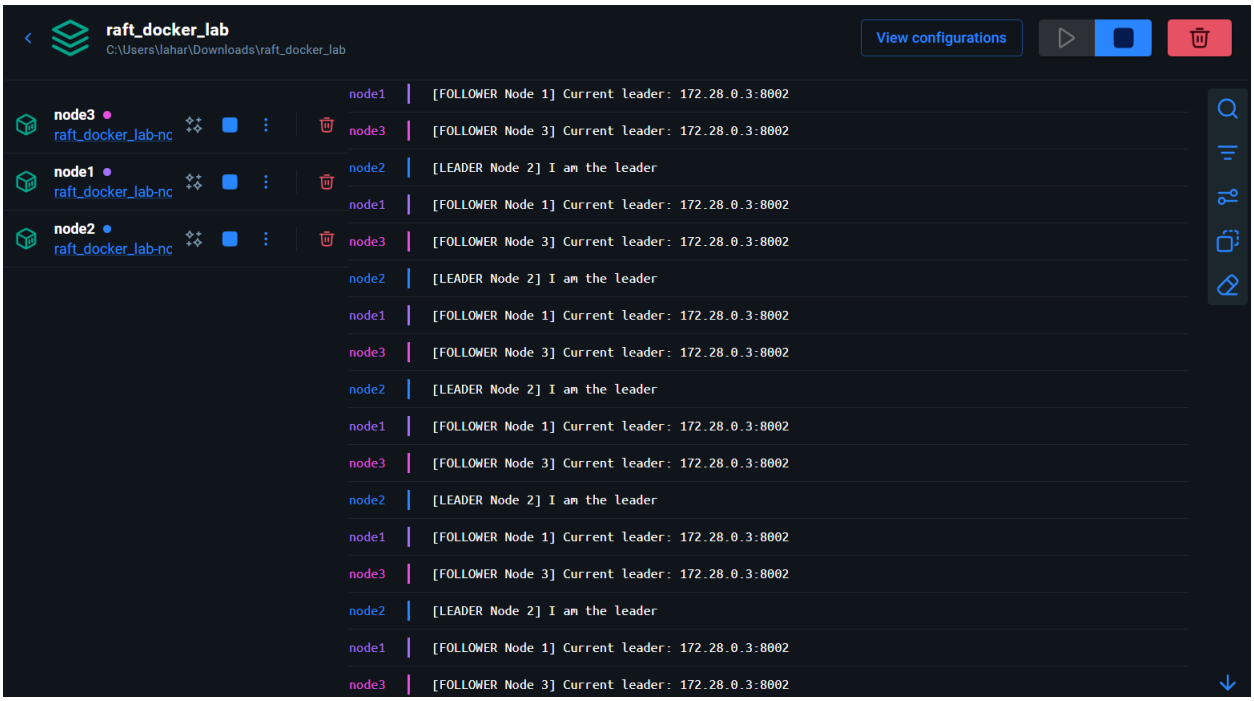
SS 1

```
PS C:\Users\lahar\Downloads\raft_docker_lab> dir


    Directory: C:\Users\lahar\Downloads\raft_docker_lab


Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
-a----        27-03-2026     14:46            562 docker-compose.yml
-a----        27-03-2026     14:46            450 Dockerfile
-a----        27-03-2026     14:46           1377 node.py
-a----        27-03-2026     14:46              6 requirements.txt


PS C:\Users\lahar\Downloads\raft_docker_lab>
```

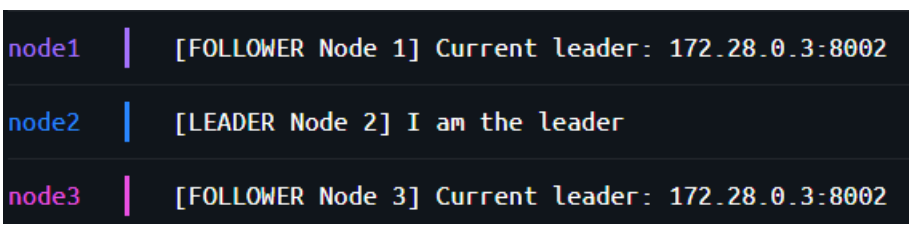
SS 2A

```
#19 [node2] resolving provenance for metadata file
#19 DONE 0.2s

#20 [node3] resolving provenance for metadata file
#20 DONE 0.1s

#21 [node1] resolving provenance for metadata file
#21 DONE 0.0s
[+] up 7/7
 ✔Image raft_docker_lab-node3     Built
 ✔Image raft_docker_lab-node1     Built
 ✔Image raft_docker_lab-node2     Built
 ✔Network raft_docker_lab_raftnet Created
 ✔Container node2                 Created
 ✔Container node1                 Created
 ✔Container node3                 Created
Attaching to node1, node2, node3
node2  | Starting Node 2 at 172.28.0.3:8002


node2  | [Node 2] Waiting for leader election...
node1  | Starting Node 1 at 172.28.0.2:8001
node1  | [Node 1] Waiting for leader election...
```

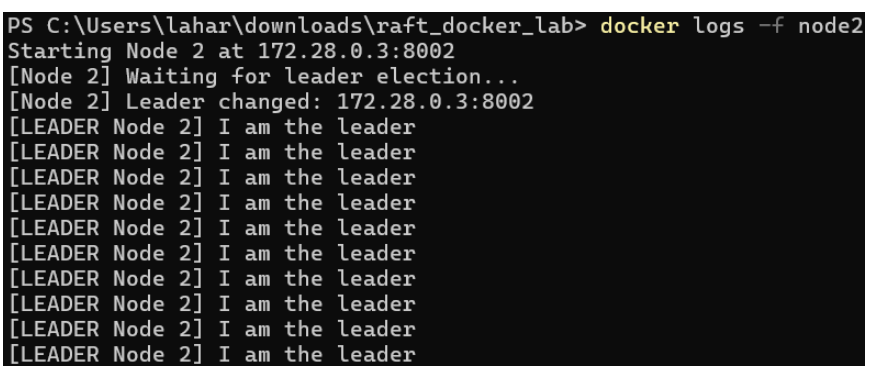
SS 2B



SS 3A



SS 3B



SS 3C

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

SS3D

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

SS 4

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED          STATUS          PORTS     NAMES
d071a88a90b9   raft_docker_lab-node3   "python node.py 3"   12 minutes ago   Up 12 minutes             node3
32998ceeed6a   raft_docker_lab-node1   "python node.py 1"   12 minutes ago   Up 12 minutes             node1
02b3fd16f016   raft_docker_lab-node2   "python node.py 2"   12 minutes ago   Up 12 minutes             node2
PS C:\Users\lahar\downloads\raft_docker_lab>
```

SS 5

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker network inspect raft_docker_lab_raftnet
[
    {
        "Name": "raft_docker_lab_raftnet",
        "Id": "1959db9e24b0d6cba682a10f12b423f1dcc140744072832f31708bd002a286fe",
        "Created": "2026-03-27T09:28:39.224960486Z",
        "Scope": "local",
        "Driver": "bridge",
        "EnableIPv4": true,
        "EnableIPv6": false,
        "IPAM": {
            "Driver": "default",
            "Options": null,
            "Config": [
                {
                    "Subnet": "172.28.0.0/16",
                    "Gateway": "172.28.0.1"
                }
            ]
        },
        "Internal": false,
        "Attachable": false,
        "Ingress": false,
        "ConfigFrom": {
            "Network": ""
        },
        "ConfigOnly": false,
        "Options": {
            "com.docker.network.enable_ipv4": "true",
            "com.docker.network.enable_ipv6": "false"
        },
        "Labels": {
            "com.docker.compose.config-hash": "7b955128463677559d4edba13e9fe8f69dd803b2dfc190dd0f2333946fc6a5d0",
            "com.docker.compose.network": "raftnet",
            "com.docker.compose.project": "raft_docker_lab",
            "com.docker.compose.version": "5.1.0"
        },
        "Containers": {
            "02b3fd16f0160772de59433b5ac1a3c888acca388ef56956ab1cdaee77df9267": {
                "Name": "node2",
                "EndpointID": "60162e8768f5e6a463a91d964b8e88deff47700a4f4ed8cc545bf6fc62b674ed",
```

SS 6A

```
node1    [Node 1] Leader changed: None
         [Node 1] Waiting for leader election...

node3    [FOLLOWER Node 3] Current leader: 172.28.0.3:8002

node1    [Node 1] Waiting for leader election...

node3    [Node 3] Leader changed: None
         [Node 3] Waiting for leader election...

node1    [Node 1] Leader changed: 172.28.0.4:8003
         [FOLLOWER Node 1] Current leader: 172.28.0.4:8003

node3    [Node 3] Leader changed: 172.28.0.4:8003
         [LEADER Node 3] I am the leader
```

SS 6B

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
[FOLLOWER Node 1] Current leader: 172.28.0.3:8002
```

SS 6C

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
[FOLLOWER Node 3] Current leader: 172.28.0.3:8002
```

SS 7A

```
node2 | [FOLLOWER Node 2] Current leader: 172.28.0.4:8003
node1 | [FOLLOWER Node 1] Current leader: 172.28.0.4:8003
node3 | [LEADER Node 3] I am the leader
```

SS 7B

```
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
[FOLLOWER Node 2] Current leader: 172.28.0.4:8003
```

SS 8

```
PS C:\Users\lahar\Downloads\raft_docker_lab> docker ps
CONTAINER ID   IMAGE                  COMMAND              CREATED         STATUS         PORTS     NAMES
9eeb9d4bc5aa   raft_docker_lab-node1  "python node.py 1"   6 minutes ago   Up 6 minutes             node1
08144f3b13d3   raft_docker_lab-node2  "python node.py 2"   6 minutes ago   Up 6 minutes             node2
eb2808260a0c   raft_docker_lab-node4  "python node.py 4"   6 minutes ago   Up 6 minutes             node4
698bc542be2e   raft_docker_lab-node3  "python node.py 3"   6 minutes ago   Up 6 minutes             node3
52723f725278   raft_docker_lab-node5  "python node.py 5"   6 minutes ago   Up 6 minutes             node5
PS C:\Users\lahar\Downloads\raft_docker_lab>
```

SS 9

```
PS C:\Users\lahar\Downloads\raft_docker_lab> docker logs -f node1
Starting Node 1 at 172.28.0.2:8001
[Node 1] Waiting for leader election...
[Node 1] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
[FOLLOWER Node 1] Current leader: 172.28.0.5:8004
```

SS 10

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node2
Starting Node 2 at 172.28.0.3:8002
[Node 2] Waiting for leader election...
[Node 2] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
[FOLLOWER Node 2] Current leader: 172.28.0.5:8004
```

SS 11

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node3
Starting Node 3 at 172.28.0.4:8003
[Node 3] Waiting for leader election...
[Node 3] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
[FOLLOWER Node 3] Current leader: 172.28.0.5:8004
```

SS 12

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node4
Starting Node 4 at 172.28.0.5:8004
[Node 4] Waiting for leader election...
[Node 4] Leader changed: 172.28.0.5:8004
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
[LEADER Node 4] I am the leader
```

SS 13

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker logs -f node5
Starting Node 5 at 172.28.0.6:8005
[Node 5] Waiting for leader election...
[Node 5] Leader changed: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
[FOLLOWER Node 5] Current leader: 172.28.0.5:8004
```

SS 14

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker stop node1
node1
PS C:\Users\lahar\downloads\raft_docker_lab> docker stop node2
node2
PS C:\Users\lahar\downloads\raft_docker_lab> docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED          STATUS          PORTS     NAMES
eb2808260a0c   raft_docker_lab-node4   "python node.py 4"   12 minutes ago   Up 12 minutes             node4
698bc542be2e   raft_docker_lab-node3   "python node.py 3"   12 minutes ago   Up 12 minutes             node3
52723f725278   raft_docker_lab-node5   "python node.py 5"   12 minutes ago   Up 12 minutes             node5
PS C:\Users\lahar\downloads\raft_docker_lab>
```

SS15

```
PS C:\Users\lahar\downloads\raft_docker_lab> docker stop node3
node3
PS C:\Users\lahar\downloads\raft_docker_lab> docker ps
CONTAINER ID   IMAGE                   COMMAND              CREATED          STATUS          PORTS     NAMES
eb2808260a0c   raft_docker_lab-node4   "python node.py 4"   13 minutes ago   Up 13 minutes             node4
52723f725278   raft_docker_lab-node5   "python node.py 5"   13 minutes ago   Up 13 minutes             node5
PS C:\Users\lahar\downloads\raft_docker_lab>
```

SS 16

```
[Node 4] Leader changed: None
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
[Node 4] Waiting for leader election...
```

SS 17

```
[Node 5] Leader changed: None
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
[Node 5] Waiting for leader election...
```

Q.Why did leader election not occur when only 2 nodes were active in a 5-node cluster?

Leader election did not occur because the Raft consensus algorithm requires a majority (quorum) of nodes to elect a leader, and with only 2 nodes active in a 5-node cluster, this condition is not satisfied.